

Problem 1.2 — The VAE Loss Function

Background and Objective

A **Variational Autoencoder** (VAE) maps each input to a *distribution* in latent space ($\mu, \log \sigma^2$) rather than a point, then decodes a sample drawn from that distribution. Training minimises two objectives: (1) **Reconstruction Loss** — how faithfully the decoder reproduces the input, and (2) **KL Divergence** — a regulariser pulling the latent distribution toward $\mathcal{N}(0, I)$. The goal here is to implement this combined loss, train on MNIST for 20 epochs, plot all three loss components, and analyse the effect of β on reconstruction sharpness.

Loss Function Formulation

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{KLD}}$$

Reconstruction Loss — Binary Cross-Entropy (BCE)

$$\mathcal{L}_{\text{recon}} = -\frac{1}{N} \sum_{i=1}^N \sum_{d=1}^D [x_{id} \log \hat{x}_{id} + (1 - x_{id}) \log(1 - \hat{x}_{id})]$$

x_{id} : true pixel, $\hat{x}_{id}$: decoder output, N : batch size, $D = 784$.

KL Divergence

$$\mathcal{L}_{\text{KLD}} = -\frac{1}{2N} \sum_{i=1}^N \sum_{j=1}^d (1 + \log \sigma_{ij}^2 - \mu_{ij}^2 - \sigma_{ij}^2)$$

Role of β

$\beta = 1$: original VAE [1]. $\beta > 1$ (β -VAE): more disentangled latent space, lower reconstruction quality.
 $\beta < 1$: sharper reconstructions, less structured latent space.

Model Architecture

Fully connected (no convolutions). Latent dim: 20. Optimiser: Adam (10^{-3}). Batch: 128.

Component	Layer	Dimensions
Encoder	FC + ReLU	784 $\rightarrow$ 400
	FC (mean / log-var)	400 $\rightarrow$ 20 each
Reparam.	$z = \mu + \varepsilon \cdot \sigma, \varepsilon \sim \mathcal{N}(0, I)$	dim 20
Decoder	FC + ReLU, FC + Sigmoid	20 $\rightarrow$ 400 $\rightarrow$ 784

Results — Baseline ($\beta = 1.0$)

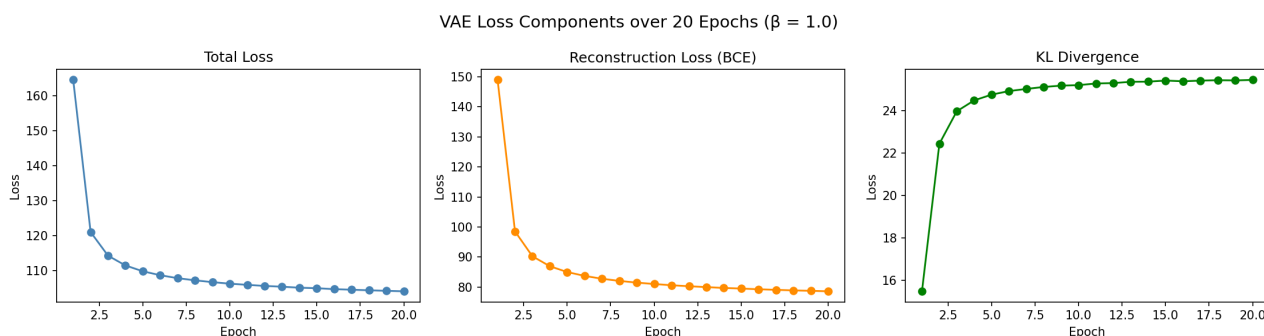
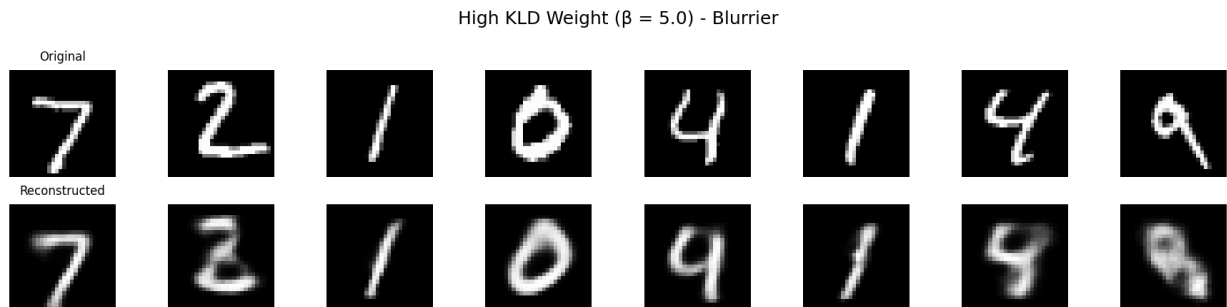
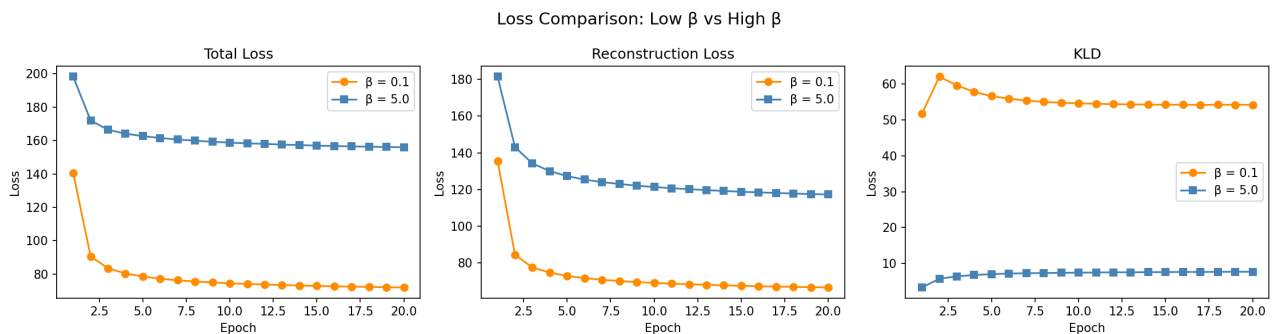


Figure 1: Total, Reconstruction (BCE), and KL Divergence over 20 epochs ($\beta = 1.0$).

Epoch	Total	Recon	KLD	Notes
1	164.45	148.98	15.47	Fast initial drop
5	109.73	84.99	24.74	KLD plateauing
10	106.19	81.00	25.19	Slow convergence
20	104.01	78.58	25.43	Final values

Key observations:

- Total and Recon losses drop sharply in epochs 1–2, then converge slowly.
- KLD *rises* from ≈ 15 to a plateau near 25 — the encoder gradually uses more latent capacity as reconstruction pressure increases.

Effect of β on Reconstruction Sharpness(a) $\beta = 0.1$ — low KLD weight. Top: originals. Bottom: reconstructions.(b) $\beta = 5.0$ — high KLD weight. Top: originals. Bottom: reconstructions.Figure 2: Reconstruction quality: low vs. high β .Figure 3: Loss curves: $\beta = 0.1$ (orange) vs. $\beta = 5.0$ (blue) over 20 epochs.**Numerical summary at epoch 20**

Model	Total	Recon	KLD
$\beta = 0.1$	72.05	66.63	54.20
$\beta = 1.0$ (baseline)	104.01	78.58	25.43
$\beta = 5.0$	156.03	117.37	7.73

Observation and Analysis

What happens to sharpness when β is increased?

β	Recon Loss	KLD	Image quality
Low (0.1)	Lower (≈ 67)	Higher (≈ 54)	Sharper, crisper digits
High (5.0)	Higher (≈ 117)	Lower (≈ 8)	Blurry, detail lost

Low β : Reconstruction dominates. The encoder places latent codes freely in $\mathbb{R}^{20}$, giving the decoder rich, precise information — producing sharp outputs. High KLD (≈ 54) confirms the posterior deviates from $\mathcal{N}(0, I)$.

High β : KLD dominates, forcing all inputs toward $\mathcal{N}(0, I)$. The compressed latent code carries little input-specific detail, so the decoder produces blurry, averaged reconstructions. Low KLD (≈ 8) confirms tight regularisation.

The Fundamental VAE Trade-off

$$\underbrace{\text{Reconstruction fidelity}}_{\text{lower } \beta} \longleftrightarrow \underbrace{\text{Latent-space regularity}}_{\text{higher } \beta}$$

A well-regularised latent space benefits *generation* and disentanglement but hurts *reconstruction sharpness*. The choice of β depends on the downstream task.

Summary

1. VAE loss = BCE reconstruction + β ·KLD regularisation.
2. At $\beta = 1$: total loss $164 \rightarrow 104$; KLD rises $15 \rightarrow 25$; Recon $149 \rightarrow 79$.
3. Higher $\beta \Rightarrow$ lower KLD, higher recon error, *blurrier* images.
4. Lower $\beta \Rightarrow$ higher KLD, lower recon error, *sharper* images.

References

- [1] D. P. Kingma and M. Welling, “Auto-Encoding Variational Bayes,” *ICLR*, 2014.
- [2] I. Higgins et al., “ β -VAE: Learning Basic Visual Concepts...,” *ICLR*, 2017.